THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Department of Software Engineering

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STUDENT ROLL NO:	**2024-SE-15**
LAB:	**OPEN ENDED LAB**
COURSE TITLE:	**Computer Networks**
COURSE CODE:	**CS-2203**
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LAB: OPEN ENDED-ENTERPRISE NETWORK DESIGN

1. OBJECTIVE AND NETWORK REQUIREMENTS

The objective of this lab is to design and implement a secure, segmented campus network for two departments using a single physical router interface.

- **Hardware:** 1× Cisco 2911 Router (R1), 2× Cisco 2960 Switches (SW1, SW2).
- **End Devices:** 11 PCs total (5 for Administration, 6 for Academic).

2. ARCHITECTURAL CHOICES & JUSTIFICATIONS

The Extended Star Topology utilizing a Router-on-a-Stick architecture was selected as the preferred design for this network. This approach was chosen and configured based on the following justifications:

- **ROUTER PORT CONSERVATION:** By daisy-chaining the switches and connecting them to the router via a single 802.1Q trunk link, the design preserves limited physical Layer 3 interfaces for future expansion (such as WAN or server links).
- **VLAN SCALABILITY:** Utilizing an 802.1Q trunk across the switch backbone allows traffic from multiple logical networks (VLANs 10 and 20) to securely traverse shared links, meaning end devices can be placed on any switch regardless of their department.
- **EFFICIENT RESOURCE MANAGEMENT:** A centralized DHCP server on the router automates IP provisioning, while Variable Length Subnet Masking (VLSM with /28 subnets) prevents the address exhaustion common with default /24 allocations.

3. IP ADDRESSING SCHEME (VLSM)

Department	VLAN	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Administration	10	192.168.10.0/28	255.255.255.240	192.168.10.2 – .14	192.168.10.1
Academic	20	192.168.10.16/28	255.255.255.240	192.168.10.18 – .30	192.168.10.17

4. SWITCH PORT ASSIGNMENT

Switch	Physical Port	Connected Device	Port Mode	Assigned VLAN
SW1	Fa0/1 - Fa0/4	Admin PCs 1–4	Access	VLAN 10
SW1	Fa0/6	Academic PC 1	Access	VLAN 20
SW1	Gi0/1, Gi0/2	R1, SW2	Trunk	802.1Q (All)
SW2	Fa0/2	Admin PC 5	Access	VLAN 10
SW2	Fa0/3 - Fa0/7	Academic PCs 2–6	Access	VLAN 20
SW2	Gi0/1	SW1	Trunk	802.1Q (All)

5. IMPLEMENTATION PROCEDURE

PHASE 1: DEVICE PLACEMENT & CABLING

- Placed 1× Cisco 2911 router, 2× 2960 switches, and 11 PCs onto the workspace.
- Connected PCs to switch access ports and SW1 to R1 using Copper Straight-Through cables.
- Connected SW1 to SW2 using a Copper Cross-Over cable to establish the backbone switch-to-switch link.

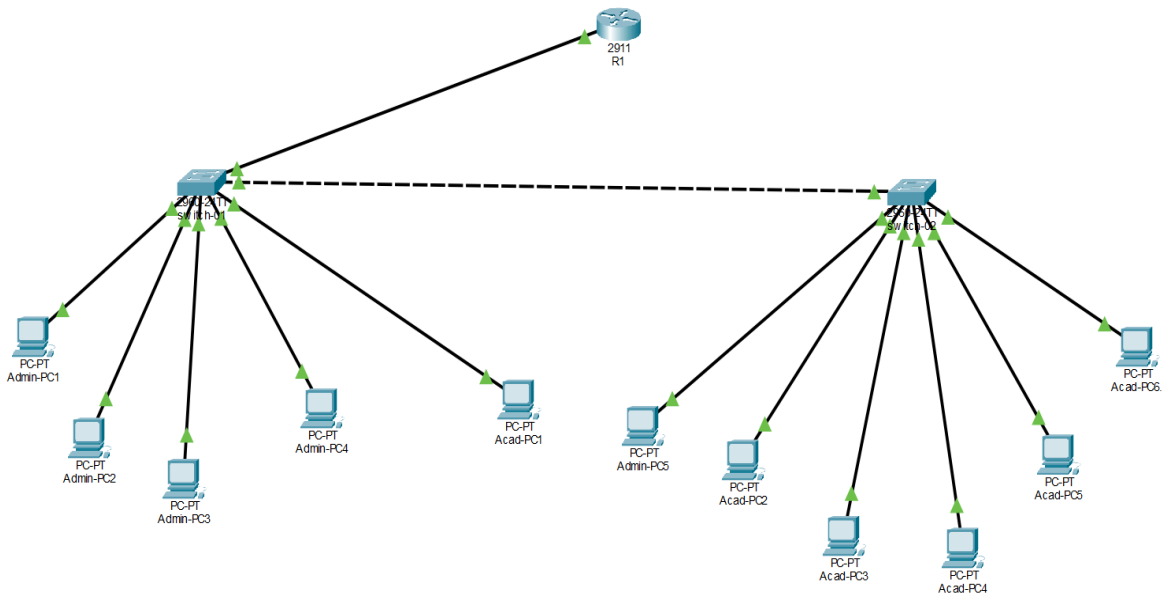


Figure 1: Logical Diagram of Network Topology

PHASE 2: SWITCH CONFIGURATION

- Created VLAN 10 (Administration) and VLAN 20 (Academic) globally on both SW1 and SW2.
- Configured the designated FastEthernet interfaces as access ports and mapped them to their respective departmental VLANs.

```
Switch(config)#hostname SW1
SW1(config)#
SW1(config)#vlan 10
SW1(config-vlan)# name ADMIN
SW1(config-vlan)#vlan 20
SW1(config-vlan)# name ACADEMIC
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#interface range fastEthernet 0/1 - 4
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config-if-range)#exit
SW1(config)#
SW1(config)#interface fastEthernet 0/6
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 20
SW1(config-if)#exit
SW1(config)#
SW1(config)#interface range gigabitEthernet 0/1 - 2
SW1(config-if-range)# switchport mode trunk

SW1(config-if-range)#exit
```

Figure 2: Switch01 configuration

- Configured the GigabitEthernet uplinks between the switches and the core router as 802.1Q trunk ports to pass multi-VLAN traffic.

```
Switch(config)#hostname SW2
SW2(config)#
SW2(config)#vlan 10
SW2(config-vlan)# name ADMIN
SW2(config-vlan)#vlan 20
SW2(config-vlan)# name ACADEMIC
SW2(config-vlan)#exit
SW2(config)#
SW2(config)#interface fastEthernet 0/2
SW2(config-if)# switchport mode access
SW2(config-if)# switchport access vlan 10
SW2(config-if)#exit
SW2(config)#
SW2(config)#interface range fastEthernet 0/3 - 7
SW2(config-if-range)# switchport mode access
SW2(config-if-range)# switchport access vlan 20
SW2(config-if-range)#exit
SW2(config)#
SW2(config)#interface gigabitEthernet 0/1
SW2(config-if)# switchport mode trunk
SW2(config-if)#exit
```

Figure 3: Configuration of Switch 02

PHASE 3: ROUTER & DHCP CONFIGURATION

- Activated the physical Gi0/0 router interface without assigning a base IP address.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#interface gigabitEthernet 0/0
R1(config-if)# no shutdown

R1(config-if)# exit
R1(config)#
R1(config)#interface gigabitEthernet 0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.240
R1(config-subif)# exit
R1(config)#
R1(config)#interface gigabitEthernet 0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.10.17 255.255.255.240
R1(config-subif)# exit
R1(config)#
R1(config)#ip dhcp excluded-address 192.168.10.1
R1(config)#ip dhcp excluded-address 192.168.10.17
R1(config)#
R1(config)#ip dhcp pool ADMIN
R1(dhcp-config)# network 192.168.10.0 255.255.255.240
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# exit
R1(config)#
R1(config)#ip dhcp pool ACADEMIC
R1(dhcp-config)# network 192.168.10.16 255.255.255.240
R1(dhcp-config)# default-router 192.168.10.17
R1(dhcp-config)# exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

Figure 4: Router Configuration

- Created two sub-interfaces (Gi0/0.10 and Gi0/0.20), configured them with 802.1Q encapsulation, and assigned their respective default gateway IPs.
- Configured two distinct DHCP pools on the router to dynamically assign IP addresses, subnet masks, and gateways to the end devices.
- Explicitly excluded the router gateway IPs from the DHCP distribution pools to prevent IP conflicts.

6. TESTING & VERIFICATION

- **DHCP Verification:** Checked end devices to confirm all 11 PCs successfully leased dynamic IP configurations matching their physical VLAN assignments.
- **Reachability Test:** Executed successful ICMP ping requests from the Administration network (VLAN 10) to the Academic network (VLAN 20), proving that Inter-VLAN routing, sub-interfaces, and trunking were fully operational.

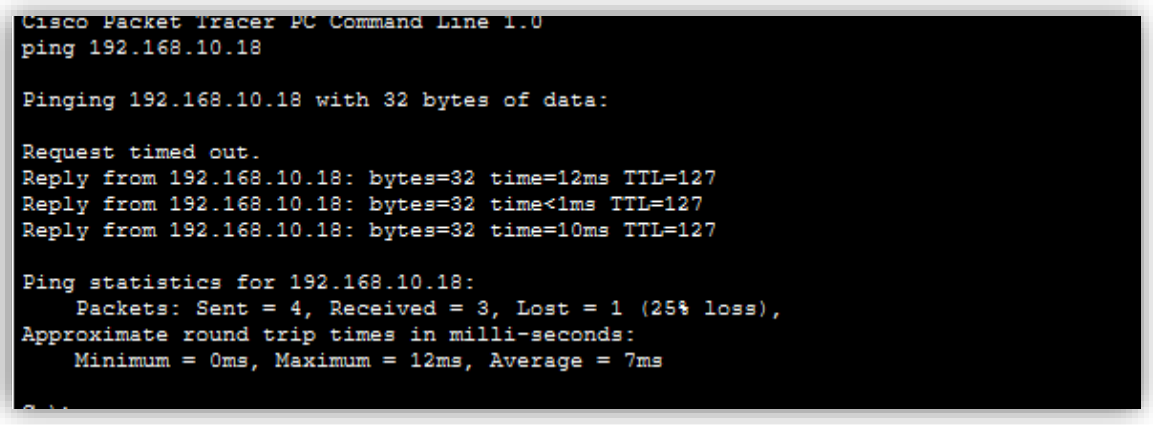


Figure 5: Testing of intra department using ping

7. FUTURE IMPROVEMENTS

- **Extended Access Control Lists (ACLs):** Implement security policies on the router to prevent unauthorized Academic network traffic from reaching sensitive Administration resources.
- **Gateway Redundancy:** Deploy a secondary core router utilizing Hot Standby Router Protocol (HSRP) to eliminate the single point of failure at the gateway.